

Einsendaufgaben – Aufgabe 3.4

61211 Analysis

Aufgabe 3.4

a)

$$\frac{\partial f(r)}{\partial x} = \frac{\partial f(\sqrt{x^2 + y^2 + z^2})}{\partial x} = 2x \cdot \frac{1}{2r} \cdot f'(r) = f'(r) \cdot \frac{x}{r}$$

$$\frac{\partial f(r)}{\partial y} = f'(r) \cdot \frac{y}{r}$$

$$\frac{\partial f(r)}{\partial z} = f'(r) \cdot \frac{z}{r}$$

$$\Rightarrow \text{grad } f(r) = f'(r) \cdot \frac{(x, y, z)}{r}$$

b)

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial f'(r) \cdot \frac{x}{r}}{\partial x} = \frac{r \left(\frac{x^2}{r} f''(r) + f'(r) \right) - f'(r) \cdot \frac{x^2}{r}}{r}$$

$$= \frac{x^2 f''(r)}{r^2} + \frac{f'(r)}{r} - \frac{f'(r) \cdot x^2}{r^3}$$

$$\begin{aligned} \Delta f(r) &= \frac{f''(r)}{r^2} (x^2 + y^2 + z^2) + 3 \frac{f'(r)}{r} - \frac{f'(r)}{r^3} (x^2 + y^2 + z^2) \\ &= f''(r) + 3 \frac{f'(r)}{r} - \frac{f'(r)}{r} \\ &= f''(r) + 2 \frac{f'(r)}{r} \end{aligned}$$

c)

$$f : (0, \infty) \rightarrow \mathbb{R}, \quad x \mapsto \frac{1}{x}$$

$$f'(x) = -\frac{1}{x^2}$$

$$f''(x) = \frac{2}{x^3}$$

Nach b) folgt

$$\begin{aligned}\Delta\left(\frac{1}{r}\right) &= \Delta f(r) = f''(r) + \frac{2}{r}f'(r) \\ &= \frac{2}{r^3} + \frac{2}{r}\left(-\frac{1}{r^2}\right) \\ &= 0\end{aligned}$$